Q.26 What are the causes of vibrations?

Q.27 Differentiate between two-stroke and four-stroke engine.

Q.28 Explain the concept of static & dynamic balancing

Q.29 What is clutch? Write its various functions.

Q.30 Explain the different types of Followers.

Q.31 Distinguish between disc brake with drum brake.

Q.32 Write about the terms: free vibrations, forced vibrations and damped vibrations.

Q.33 Explain the terms Periodic motion, Time period, Frequency, Amplitude of motion and cycle.

Q.34 Explain gear drive and its types.

Q.35 Define crowning of pulley.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain working of Otto cycle by drawing on p-v and T-s diagrams and derive an expression for finding out the thermal efficiency.

Q.37 A horizontal cross compound steam engine of 300 kW runs at 100 rpm. The speed of the engine is to be maintained within 0.5% of the mean speed. The flywheel has a mass 2000 kg with radius of gyration of 1m. Determine the co-efficient of fluctuation of energy.

Q.38 What is the function of Brake. Explain any one type of brake in detail.

No. of Printed Pages : 4 181845/121845/032343

Roll No. ....

4th Sem / T & D, Prod

Subject:-Basics of Mechanical Engineering

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 What is thermodynamics ?

a) Study of the relationship between heat and other forms of energy

b) Study of the conversion of chemical energy to other forms of energy

c) Study of the relationship between mechanical energy to other forms of energy

d) Study of the conversion of mechanical energy to other forms of energy view answer

Q.2 In a two-stroke engine, the four functions performed in SI engine are done in which two strokes?

a) Expansion stroke and compression stroke

b) Intake stroke and exhaust stroke

c) Compression stroke and power stroke

d) Compression stroke and expansion stroke

Q.3 The coefficient of fluctuation of energy =

a) Maximum fluctuation of energy / work done per cycle

b) Fluctuation of energy / Work done per cycle

c) Maximum fluctuation of energy/ Mean speed

d) Fluctuation of energy / Mean speed

Q.4 The angle between the axis of the follower and the normal to the pitch curve is known as the

a) Base angle b) Pressure angle
c) Pitch angle d) Prime angle

Q.5 Which of the following is not correct in respect of "V" belt drive

a) It is compact
b) It can be easily replaced and maintenance is easy
c) It causes less noise and vibration
d) It is used where distance between driver and driven pulley is more

Q.6 Which is the effect of the slip of the belt on the velocity ratio of belt drive?

a) Increases
b) Decreases
c) May increase or decrease
d) Remains constant

Q.7 The following is not a friction clutch

a) Fluid clutch b) Centrifugal clutch
c) Cone clutch d) Disc clutch

Q.8 In which of the following brakes the force acting on the brake drum is in axial direction?

a) Block brake b) Shoe brake
c) band brake d) Cone brake

Q.9 When a body is subjected to transverse vibration, the stress induced in a body will be

a) Shear stress b) Compressive stress
c) Tensile stress d) None of the mentioned

Q.10 Often an unbalance of forces is produced in rotary or reciprocating machinery due to the__________

a) Centripetal forces b) Centrifugal forces
c) Friction forces d) Inertia forces

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define Zeroth law of thermodynamics
Q.12 Define SI engine
Q.13 Define Fluctuation of energy
Q.14 Define Radial or disc cam
Q.15 Define Piston
Q.16 Define Velocity ratio
Q.17 Define Cross belt drive
Q.18 Define Static friction.
Q.19 Define Dynamic balance
Q.20 Define Resonance

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are the main advantages of I.C. engines?
Q.22 Explain First law of thermodynamics.
Q.23 What do you understand by the terms "Thermal Efficiency" and "Mechanical Efficiency"?
Q.24 Write the types of flywheels. Which type of flywheel is preferred ?
Q.25 Derive a relation between fluctuation of energy and co-efficient of speed.